

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	SWTID-2026-1307
Project Title	Script-Controlled ACL-Restrict Record Access Based on Field Value.
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to secure sensitive records on the ServiceNow platform using script-controlled Access Control Lists (ACLs) that decide access based on the field value of each record. It tackles the gap where role-based access alone exposes confidential records, promising better data protection, reduced risk of data leaks and audit-ready security. Key features include a scripted record-level ACL and server-side enforcement across forms, lists, reports and APIs.

Project Overview	
Objective	The primary objective is to restrict access to records based on the value of a field (for example, Sensitivity = Confidential) by using a script-controlled ACL, so that only authorized users can read or modify such records.
Scope	The project designs, scripts and tests record-level ACLs (read / write) on a ServiceNow table, covering role, owner, assignment group and department checks, and validates them by impersonating different users. It does not cover external identity systems or third-party plugins.
Problem Statement	
Description	Default role-based ACLs grant access to the whole table and cannot check the value inside a record. Hiding data using UI policies or client scripts is not real security and can be bypassed through lists, reports and APIs. As a result, confidential records are visible to unauthorized users.
Impact	Solving this will protect confidential records, prevent data leaks, meet audit and compliance needs, and increase user trust in the platform, contributing to organizational security and success.
Proposed Solution	
Approach	Create a script-controlled ACL on the target table. The ACL script reads the record's field value (current.field) and compares it with the logged-in user (role, user ID, group, department) using server-side JavaScript. If the conditions match, answer = true; otherwise access is denied.

Key Features	• Scripted record-level ACL for read and write operations. • Access decision based on the field value of the record. • Server-side enforcement in forms, lists, reports, search and REST API. • Access for owner, assigned group, authorized role and admin only. • Testing by impersonating users with different roles.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5 / equivalent, 4 cores (no GPU required)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Platform	Cloud platform for building ACLs	ServiceNow Personal Developer Instance (PDI)
Scripting	Language used in ACL script	Server-side JavaScript (GlideRecord, GlideSystem: gs.hasRole, gs.getUserID)
Development Environment	Tools / IDE	ServiceNow Studio, Script Editor, Chrome / Edge browser, Impersonate User, ACL debugging (Security Debugger)
Data		
Data	Source, size, format	Sample records created in ServiceNow (custom table), about 50 to 100 records; test users and roles; format: table records / CSV import